Information Processing and Management 61 (2024) 103616

Contents lists available at ScienceDirect

Information Processing and Management

journal homepage: www.elsevier.com/locate/ipm

Confidence-based Syntax encoding network for better ancient Chinese understanding

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ARTICLE INFO

Keywords:
Ancient Chinese understanding
Syntax encoding
Syntax parse confidence
Ancient poetry thematic classification
Ancient–modern Chinese translation

ABSTRACT

While neural-based models continue to make rapid strides, syntax remains a foundational element in the domain of Natural Language Processing (NLP), particularly in the context of Chinese language understanding. However, there exists a significant gap in research that integrates syntactic information for the understanding of ancient Chinese, primarily due to the lack of high-quality syntactic annotations. This paper explores the untapped potential of syntax to enhance ancient Chinese understanding, leveraging the "not-so-perfect" noisy syntax trees generated by unsupervised derivations and modern Chinese syntax parsers. To achieve this, we introduce a novel syntax encoding component: the confidence-based syntax encoding network (cSEN). This component is tailored to mitigate the side-effects arising from the noise associated with unsupervised syntax derivations and the incompatibility between ancient and modern Chinese. We validate the importance of syntax information and the efficacy of our cSEN through experimental tasks, specifically ancient poetry theme classification and ancient–modern Chinese translation. Our findings suggest that proper implementation of syntactic information can effectively enhance model understanding of ancient Chinese. The introduced cSEN proves vital in noise-rich environments, potentially revolutionizing the way information professionals approach and utilize ancient Chinese texts.

1. Introduction

Chinese, with its 6000 year history, is a reservoir of profound cultural heritage (Norman, 1988). Ancient Chinese literature, from classical poetry to historical narratives, signifies the core of this heritage. These texts, rich in cultural content, open doors to extensive linguistic and cultural studies (Zhang & Zhang, 2015). Over its long history, the Chinese language has experienced significant evolutions, like the shift from classical to vernacular styles in the early 20th century (Weiping, 2017). This evolution created distinct variations between ancient and modern Chinese, necessitating tailored NLP strategies. These approaches should consider both the unique linguistic attributes of ancient Chinese (Feng & Li, 2023; Song, 2022) and the historical–cultural contexts of their origins (Chang, Shiue, Yeh, & Demberg, 2021). Thus, studying ancient Chinese merges linguistic challenges with a deep dive

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https://doi.org/10.1016/j.ipm.2023.103616
Received 4 July 2023; Received in revised form 11 December 2023; Accepted 17 December 2023
Available online 29 December 2023
0306-4573/

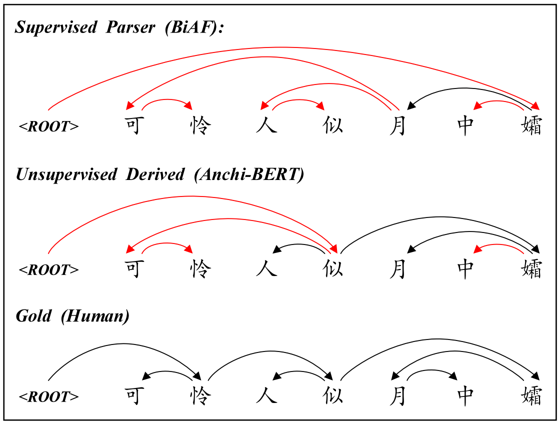

Fig. 1. Comparison of dependency parses from different parsers, where red arcs indicate prediction noises. For an English interpretation of the ancient Chinese sentence presented in this figure, please see footnote 1. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

into China's historical tapestry. To better interpret ancient Chinese using modern Natural Language Processing (NLP) technologies, numerous endeavors have been launched. These include the creation of robust datasets (Fan & Wang, 2022; Lee & Kong, 2012; Li, Qi, Sun, Yi, & Zhang, 2021; Liu & Wang, 2012; Wong & Lee, 2016), the training of expansive language models (Chang et al., 2023; Zhao, Bai, Wei, & Wu, 2023), and the pioneering of advanced modeling techniques (Feng & Li, 2023; Yu, Shao, & Li, 2022).

Among the various elements vital to understanding language, syntax, as the backbone of language structure, plays a pivotal role. Syntactic analysis, commonly referred to as parsing, breaks a sentence into its grammatical structure. Specifically, our emphasis in this work lies on "dependency parses", where words in a sentence are connected based on their grammatical and functional relationships, denoted by arrows or arcs between them. This form of parsing reveals the interdependence of words within a sentence. The incorporation of syntactic features can bridge gaps between ancient and modern interpretations, allowing models to discern relationships and capture linguistic nuances.

The effectiveness of parsed syntactic features has been validated across a wide range of NLP tasks. Their application spans domains like sentiment analysis (Zhao, Liu, Zhang, Guo, & Chen, 2021), machine reading comprehension (Guo, Guan, Li, Li, & Tan, 2020; Zhang et al., 2020), machine translation (Bugliarello & Okazaki, 2020; Currey & Heafield, 2019; Zhang, Li, Fu, & Zhang, 2019), and grammatical error correction (Li, Parnow, & Zhao, 2022). Given the success of syntactic features in these tasks and the distinct syntactic structures between modern and ancient Chinese, it is plausible that incorporating syntactic features into the processing and understanding of ancient Chinese could provide a substantial impetus for progress in this field. While syntax has proven to be effective in understanding modern Chinese (Li et al., 2018; Xia, Li, & Zhang, 2019; Zhang et al., 2020), only a few studies have effectively incorporated syntactic information into the processing of ancient Chinese texts. Most of the current works focus on explicit features, such as the temporal context of the text (era) (Chang et al., 2021) and imagery (Shen et al., 2019), while largely overlooking the importance of implicit syntactic features due to the absence of accurate syntax parsing of ancient Chinese. A recent investigation (Chen & Shao, 2023) spotlighted this, demonstrating the limitations of modern Chinese parsers when adapting to ancient Chinese.

The challenges in accurately parsing ancient Chinese syntax arise from several aspects. Firstly, direct application of modern Chinese parsers to ancient texts often results in side effects, as the language evolution over millennia has introduced syntactic disparities between the two. Secondly, the inherent poetic nature of many ancient Chinese writings means that the language is denser and more ambiguous, leading to noise in automated parses. Thirdly, cultural and contextual nuances play a significant role in understanding the syntactic structures, which are often missing in machine-driven approaches. Consider, for instance, the ancient Chinese sentence "可怜人似月中孀".[1] As illustrated in Fig. 1, we compare the parses generated by a supervised parser, an unsupervised derivation, and a human annotator. Specifically, the supervised parser is a Biaffine dependency parser (Dozat & Manning, 2016) trained on modern Chinese. The unsupervised derivation is obtained using Anchi-BERT (Tian, Yang, Liu, & Lv, 2021), a BERT variant pretrained on ancient Chinese texts.

From the comparison, it is evident that the supervised modern Chinese parser yields subpar results, failing to capture the intricate syntactic structure of the sentence. The unsupervised ancient Chinese derivation, while performing better than its supervised modern Chinese counterpart, still leaves much to be improved, particularly with incorrect predictions on the root node and local dependencies. Such discrepancies clearly underline the incompatibility between ancient and modern Chinese, highlighting the pressing need for an effective mechanism that harnesses syntax features in ancient Chinese understanding.

[1] The ancient Chinese sentence "可怜人似月中孀" can be roughly translated as "It is pitiful like Chang'e in the moon". Please note that due to the poetic nature of ancient Chinese, the translation is a loose interpretation to capture the essence rather than a direct literal translation.

In response to these challenges and to fulfill the outlined research objectives, we introduce a novel syntactic encoding structure named the Confidence-Based Syntax Encoding Network (cSEN). In cSEN, confidence is computed via a Biaffine transformation applied to the sequence representation and the syntactic graph's derived adjacency matrix. This confidence metric empowers our model to discern beneficial syntactic features from extraneous noise. Furthermore, the ancient Chinese language, in contrast to its modern counterpart, is characterized by more concise expressions and more compact structures, where each token is highly interconnected with its preceding and following tokens. Considering this distinctive linguistic characteristic, we integrate an additional graph feature, the Left–Right Branch (LRB), into our model. This feature captures local linguistic nuances to further enhance the understanding of ancient Chinese.

We have conducted experiments on two quintessential tasks related to ancient Chinese understanding: ancient poetry thematic classification and ancient–modern Chinese translation. The results show that our proposed model outperforms powerful baselines, indicating that cSEN is capable of effectively handling the noise in derived syntax trees. To the best of our knowledge, our proposed cSEN represents the first practical application of syntactic information in ancient Chinese processing. Key contributions of this study can be summarized as follows:

1. We address a critical research gap by exploring the role of syntax in ancient Chinese understanding. Our findings highlight that properly handled noisy parses can effectively enhance ancient Chinese understanding.
2. We introduce a novel architectural design — the Confidence-based Syntax Encoding Network (cSEN). This network mitigates the negative impact of noise in syntax parses, thus facilitating the utilization of syntactic features in ancient Chinese understanding.
3. We validate the effectiveness of cSEN across two representative ancient Chinese understanding tasks: thematic classification of ancient poetry and ancient–modern Chinese translation. Results show that our model consistently outperforms robust baseline methods in noisy scenarios.
4. We construct a new dataset for thematic classification of ancient Chinese poetry. Comprising 22,360 poems across 10 theme categories, this dataset paves the way for future research in this area and addresses the current shortage of annotated ancient Chinese corpora.

The remainder of this paper is structured as follows: Section 2 presents a comprehensive review of existing literature relevant to our research. Section 3 provides an in-depth explanation of our proposed methodology. Subsequently, Section 4 offers a detailed presentation of the experiments conducted and their respective results. Finally, Section 5 concludes the paper, summarizing the findings and discussing potential future directions for this line of research.

2. Related work

In this section, we have surveyed related works pertinent to our study. This includes investigations into the role of syntax in Chinese NLP studies, which inform our methodological approach, as well as explorations into ancient Chinese poetry and ancient–modern Chinese translation studies, which are closely tied to our research tasks

2.1. Syntax role in Chinese understanding

Syntax is essential in the realm of Natural Language Processing (NLP), particularly when enhanced by deep learning techniques (Linzen & Baroni, 2021). Contemporary developments in Chinese understanding exhibit an increasing reliance on syntax as a vital cue across various tasks.

One major area of syntax application is Semantic Role Labeling (SRL), as demonstrated by several studies. For instance, Qiu and Zhang (2014) developed a syntax-based system to identify relation candidates from dependency trees. Li et al. (2018) and Xia et al. (2019) further attested to the impact of syntax on SRL, taking GCN and LSTM as syntax encoders, and showing that high-quality syntactic parses and syntax-aware learning frameworks can significantly enhance SRL. Innovatively, Xia, Wang, Li, Zhang, and Zhang (2020) and Zhang, Wang, and Si (2019) embedded syntactic dependencies and representations into SRL models using self-attention blocks as syntax encoder, while Munir, Zhao, and Li (2021) utilized adaptive convolution to encode syntax, providing more flexibility to existing networks.

The role of syntax extends to the domain of information extraction tasks as well. Jiang, Wu, Bu, and Hu (2018), for instance, integrated syntactic features to augment identified triplets, thereby improving Chinese entity relation extraction. The utility of syntactic features is further echoed in Sun et al. (2022)'s work, where these features are used to encapsulate depth-level structure information. This includes non-consecutive words and their relations, serving to enhance the recognition of Chinese implicit inter-sentence relations. In the pursuit of improving named entity recognition, Zhu et al. (2022) employed syntactic dependency information to delineate entity boundaries more precisely. Meanwhile, Li, Yan, Yang, and Ma (2022) proposed a novel syntactic dependency-guided model, BERT-BiLSTM-GAM-CRF. This model leverages dependency relations to guide self-attention transformations within BERT layers, bringing performance gain in Chinese named entity recognition.

Indeed, the utility of syntax in Chinese NLP tasks extends beyond semantic role labeling and information extraction. Its application is evident in machine reading comprehension (Zhang et al., 2020), sentiment classification (Liu, Tang, & Ding, 2022; Zhang, Hu, Zhu, Jin, & Li, 2021), Chinese–English translation (Zhang, Li, Fu et al., 2019), and even in graph-based text generation (Guo, Qiu, Xue, & Zhang, 2021).

However, despite the burgeoning interest in syntax within modern Chinese understanding, its application in ancient Chinese understanding remains relatively unexplored. Historically, a few efforts have been made in this regard. These include the construction of a dependency treebank of classical Chinese poems (Lee & Kong, 2012) and the Chinese Buddhist canon (Wong & Lee, 2016). More recently, Song (2022) suggested the enhancement of Chinese couplet generation by leveraging syntactic information, using an attention module to encode Part-of-Speech (POS) information and dependency information associated with a Chinese character. Additionally, Chen and Shao (2023) detailed semantic dependency labeling rules for various special sentence patterns in ancient Chinese, comparing these with results from an automatic semantic parser.

2.2. Ancient–modern Chinese translation

Although both ancient and modern Chinese use the same characters, much like bilingual translation tasks such as Chinese–English, translating between the two forms of the language can be equally challenging, even for native speakers. This challenge stems from two principal factors: firstly, the syntactic structure and grammatical order of ancient Chinese differ substantially from those of modern Chinese (James, 1985), resulting in ancient Chinese expressions that are often more concise yet simultaneously more perplexing; secondly, ancient Chinese frequently utilizes literary devices such as allusion, metaphor, and symbolic imagery to implicitly invoke sensory and emotional experiences (Kao & Mei, 1978; Peng, 1994), which complicates the task of disambiguating the author's intended message.

In response to these distinctive characteristics and the inherent complexities of both modern and ancient Chinese, some researchers have turned to statistical and rule-based methods (Che, Zhao, Wu, Zhou, & Zhang, 2019; Lin & Wang, 2007; Liu & Wang, 2012). Despite this, the majority of contemporary solutions have predominantly adopted neural machine translation (NMT) approaches, reflecting a shift towards the utilization of more sophisticated, AI-driven methodologies in language translation and understanding.

For instance, Zhang, Li, and Su (2019) proposed an unsupervised algorithm to construct sentence-aligned ancient–modern pairs and introduced an end-to-end neural model integrating a copying mechanism and local attention for translation between ancient and modern Chinese. Liu, Yang, Qu, and Lv (2019) implemented RNN-based and Transformer-based machine translation models for the same task. Considering the monolingual nature of this task, Yang, Chen, and Chen (2021) utilized pre-trained models, UNILM (Dong et al., 2019) and Guwen-BERT, to enhance performance. In recognition of the considerable evolution in the Chinese language over time, Chang et al. (2021) introduced a time-aware translation method, predicting both the translation results and their associated era. This predicted chronological feature acts as auxiliary information to bridge the linguistic gap across different eras of the Chinese language.

More recently, researchers have leaned towards large-scale language models enriched with an ancient Chinese corpus. For example, Feng and Li (2023) proposed a method to augment ancient Chinese tagging data using distant supervision, employing relabeling to mitigate noise introduced by this approach. In a similar vein, Zhao et al. (2023) constructed a large-scale sememe knowledge graph, SKGPoetry, which aids in bridging the gap between classical Chinese poetry and modern Chinese. Finally, Chang et al. (2023) proposed a GPT model, SikuGPT, based on the corpus of Siku Quanshu, which outperforms other GPT-type models aimed at processing ancient texts in tasks such as intralingual translation and text classification.

Moreover, Liu, Xiang, Xia, and Hu (2022) explores the latent semantic spaces between ancient and modern Chinese. By drawing from the principles of contrastive learning, this method optimizes the differences between the two forms of the language, paving another promising way for better understanding of ancient Chinese texts.

2.3. Classification of ancient Chinese poetry

The classification of ancient Chinese poetry serves as a foundation for higher-level tasks such as sentiment or style-controlled poetry generation (Chen et al., 2019; Shao, Shao, Wang, Wang, & Gao, 2021; Yang, Sun, Yi, & Li, 2018). Historically, this has been approached through statistical features and traditional machine learning algorithms. For instance, building on the Vector Model for poetry style identification, Yi, He, Li, Yu, and Yi (2005) applied the Naïve Bayes Categorization Method to identify traditional Chinese poetry styles. Similarly, Hou and Frank (2015) suggested a weakly supervised sentiment classification approach that generates a sentiment lexicon based on Weighted Personalized PageRank (WPPR). Shen et al. (2019) incorporated imagery features into their analysis of Tang Poetry's sentiment.

However, more recent years have witnessed a marked shift towards neural classifiers, which have achieved remarkable improvements in performance. For example, Xuan et al. (2018) developed a poetry style recognition model by stacking a genetic algorithm with a Convolutional Neural Network (CNN), while Tang, Wang, Qi, and Sun (2020) combined a CNN with a Gated Recurrent Unit (GRU) to address poetry sentiment classification.

Taking into account the emotional continuity inherent in ancient poetry, Li and Li (2018) referenced the Valence-Arousal (VA) space model and TF-IDF for poetry representation. A classifier was subsequently constructed using a neural network algorithm for automatic tagging of ancient poetry. Moreover, Ai, Yijia, Hao, and Mingyu (2021) proposed an LDA-Transformer model that effectively captures incoherence information and grasps the writing style of classical poems, demonstrating substantial performance in poetry texts. In a novel approach, Zhao et al. (2023) constructed a knowledge graph of classical Chinese poetry based on word definitions and their sememe annotations, which exhibited optimal performance on poetry theme classification.

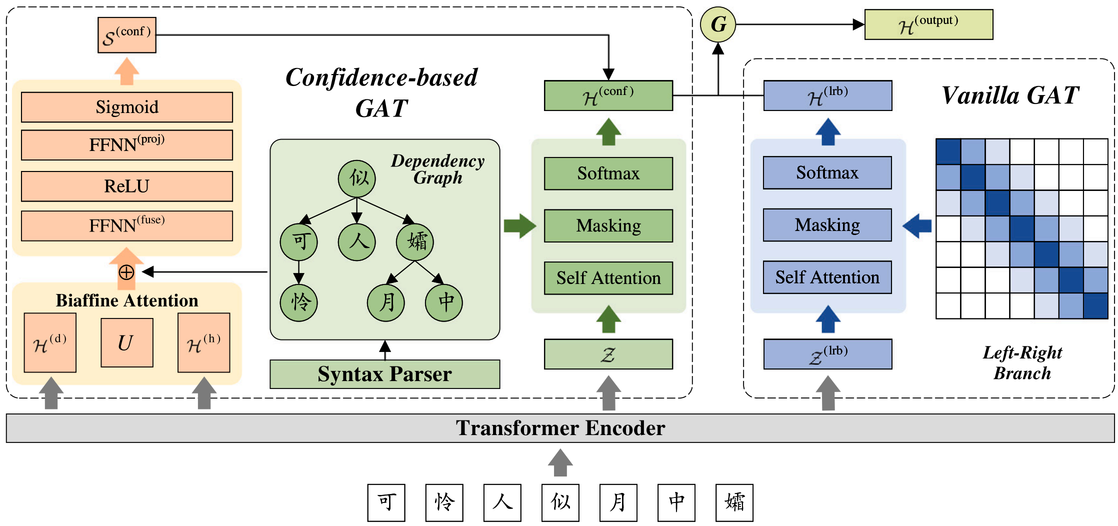

Fig. 2. Architecture of the proposed cSEN. The symbol ⊕ represents the concatenation operation, while Ⓖ denotes the gated mechanism. We depict $\mathcal{M}^{(\text{dep})}$ as a graph with arcs pointing from heads to dependencies. In $\mathcal{M}^{(\text{lrb})}$, cells are colored to emphasize local dependencies; darker shades indicate stronger correlation.

3. Model

A high-level overview of cSEN is shown in Fig. 2. At its core, cSEN leverages the strengths of Graph Attention Networks (GAT) to encode syntactic structures. To mitigate the side effect brought by noisy syntax trees, cSEN measures confidence of syntactic features using Biaffine attention and guide syntax encoding using gated fusion mechanism. Building upon this foundation, cSEN further incorporates an innovative Left–Right Branch feature which help in capturing the nuances of ancient Chinese.

In this section, we first introduce a basic Graph Attention Network (GAT) encoder, which serves as the foundational block of our model. Then, we delve into the details of our cSEN and the enhancements it brings to the table.

3.1. Vanilla GAT

The Graph Attention Network (GAT) is typically applied over a sentence encoder to abstract graph-based representations of the input text. For a given input token sequence $\mathcal{T} = t_1, t_2, \dots, t_l$, where l represents the sequence length, the output of the sentence encoder is denoted as matrix $\mathcal{H} \in \mathbb{R}^{l\times n}$, where each row $h_i \in \mathbb{R}^n$ is the representation of token t_i.

Utilizing the dependency structure of the input sequence from a syntax parser, we construct a dependency graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, wherein $\mathcal{V}$ represents the set of tokens and $\mathcal{E}$ represents the set of arcs. During the graph encoding process, the graph is represented as an adjacency matrix, denoted as $\mathcal{M}^{(\text{dep})}$. In this representation, positions corresponding to arcs and the diagonal elements are assigned a value of one, while the remaining cells are set to zero. Subsequently, linear transformation is performed by multiplying the sentence representation $\mathcal{H}$ with a matrix $\mathcal{W} \in \mathbb{R}^{n\times n'}$ for feature extraction:

$$\mathcal{Z} = \mathcal{H}\mathcal{W}, \tag{1}$$

where n' denotes the transformed feature dimension. Then, a pair-wise attention operation is performed. For every pair $t_i, t_j \in \mathcal{V}$, it concatenates the corresponding representations z_i and z_j, and applies a **LeakyReLU** activation function after taking the dot product with vector $a \in \mathbb{R}^{2n'}$:

$$\mathcal{S}^{(\text{raw})}[i,j] = \mathbf{LeakyReLU}(\left[z_i \oplus z_j\right]^T a), \tag{2}$$

where ⊕ represents the concatenation operation, and $\mathcal{S}^{(\text{raw})}$ is a score matrix of size $(l \times l)$ that captures inter-node relations. To incorporate the graph structure, the adjacency matrix $\mathcal{M}^{(\text{dep})}$ is used to constrain the function scope before applying a standard **Softmax** operation. This ensures that each token can only attend to its head tokens and itself. The obtained attention weights matrix is then employed to scale the transformed sentence representation $\mathcal{Z}$, leading to the calculation of the final attentional output:

$$\mathcal{W}^{(\text{attn})} = \mathbf{Softmax}(\mathcal{S}^{(\text{raw})} \times \mathcal{M}^{(\text{dep})}), \tag{3}$$

$$\mathcal{H}^{(\text{attn})} = \mathcal{W}^{(\text{attn})}\mathcal{Z}. \tag{4}$$

Eqs. (1)–(4) detail the computational process of a basic GAT. The attentional output acts as a syntactic-aware representation of the original sentence, where each token is contextually enhanced by its syntactic relations in the graph. This process, as a result, imbues the sentence representation with valuable syntactic information.

3.2. Confidence-based GAT

As discussed above, GAT guides the encoding process by constraining the scope of the attention computation. Therefore, any noise present in the graph will inevitably impact the encoding output. To alleviate the negative effects of noises on the model performance, we propose a confidence-based GAT. This novel approach measures the confidence of the graph adjacency matrix, enabling the model to distinguish reliable syntactic information from noises.

In alignment with the standard GAT, we begin by modeling the pair-wise relationships. Two independent linear transformations are performed separately over the sentence representation $\mathcal{H}$ to obtain the role-aware representations. The outputs, represented as $\mathcal{H}^{(\mathrm{d})}$ and $\mathcal{H}^{(\mathrm{h})}$, are of the dimension $(l \times n')$:

$$\mathcal{H}^{(\mathrm{d})} = \mathcal{H}\mathcal{W}^{(\mathrm{d})}; \mathcal{H}^{(\mathrm{h})} = \mathcal{H}\mathcal{W}^{(\mathrm{h})}. \tag{5}$$

Following this, Biaffine attention (Dozat & Manning, 2016) is calculated on the role-aware representations for pair-wise relationship scoring:

$$\mathcal{S}^{(\mathrm{bi})} = \mathcal{H}^{(\mathrm{d})} U \mathcal{H}^{(\mathrm{h})T}, \tag{6}$$

where U serves as an intermediate matrix with dimensions $(n' \times n')$. Confidence scores are then calculated by concatenating the pair-wise relationship scores and the adjacency matrix, and processing them as follows,

$$\mathcal{S}^{(\mathrm{fuse})} = \mathbf{ReLU}(\mathbf{FFNN}^{(\mathrm{fuse})}([\mathcal{S}^{(\mathrm{bi})} \oplus \mathcal{M}^{(\mathrm{dep})}])), \tag{7}$$

$$\mathcal{S}^{(\mathrm{conf})} = \mathbf{Sigmoid}(\mathbf{FFNN}^{(\mathrm{proj})}(\mathcal{S}^{(\mathrm{fuse})})), \tag{8}$$

where $\mathbf{FFNN}^{(\mathrm{fuse})}$ performs a linear transformation to fuse the two feature spaces along with a **ReLU** activation, and $\mathbf{FFNN}^{(\mathrm{proj})}$ is used to reduce the dimension from $2l$ to l, enabling the **Sigmoid** function to project the confidence features at the same magnitude as the attention scores. Using the obtained confidence scores $\mathcal{S}^{(\mathrm{conf})}$, the original attention restrain process is redefined as the following equations:

$$\mathcal{W}^{(\mathrm{conf})} = \mathbf{Softmax}(\mathcal{W}^{(\mathrm{attn})} + \mathcal{S}^{(\mathrm{conf})}), \tag{9}$$

$$\mathcal{H}^{(\mathrm{conf})} = \mathcal{W}^{(\mathrm{conf})} \mathcal{Z}. \tag{10}$$

3.3. Left–right branch feature

Inspired by the commonality of local dependencies in ancient Chinese, we introduce a novel feature, the left–right branch. This simple yet effective feature is designed to further enhance the syntax graph encoding. To model the local relationships between tokens, we populate a matrix $\mathcal{M}^{(\mathrm{lrb})}$ identical in size to $\mathcal{M}^{(\mathrm{dep})}$ as follows:

$$\mathcal{M}^{(\mathrm{lrb})}[i,j] = \begin{cases} 1, & \text{if } j \in \{i-1, i+1\} \\ 0, & \text{otherwise.} \end{cases} \tag{11}$$

This matrix implies the existence of arcs in the graph, connecting a node to its immediate left and right neighbors. The left–right branch features are encoded utilizing another GAT component, which provides a sequence representation $\mathcal{Z}^{(lrb)}$ and a positional-information-introduced attention weight matrix $\mathcal{W}^{(\mathrm{lrb})}$. The outputs from $\mathcal{M}^{(dep)}$ and $\mathcal{M}^{(lrb)}$ are amalgamated via a gated mechanism to generate the final output:

$$\mathcal{H}^{(\mathrm{lrb})} = \mathcal{W}^{(\mathrm{lrb})} \mathcal{Z}^{(\mathrm{lrb})}. \tag{12}$$

$$g = \mathbf{Sigmoid}(\mathbf{FFNN}^{(\mathrm{gate})}([\mathcal{H}^{(\mathrm{conf})} \oplus \mathcal{H}^{(\mathrm{lrb})}])), \tag{13}$$

$$\mathcal{H}^{(\mathrm{output})} = g \times \mathcal{H}^{(\mathrm{conf})} + (1-g) \times \mathcal{H}^{(\mathrm{lrb})}. \tag{14}$$

In essence, by recognizing the existence of local dependencies in ancient Chinese text and incorporating them into our model through the left–right branch feature, we refine the graph encoding process further. This refinement enhances the model's adaptability and its potential to utilize syntactic features that were not accurately predicted during the derivation phase.

To summarize, our proposed cSEN alleviates the negative effect of graph noise through a two-step process. Firstly, cSEN measures the confidence of the derived syntax parses. This confidence score is then employed to softly mask the noisy arcs and highlight the ones that were previously undetected. Secondly, in consideration of the linguistic characteristics of ancient Chinese, the Left–Right Branch feature is incorporated. This feature extends the syntax graph encoding's scope and smoothens out noise and incompatibility. The combined effect of these aspects helps alleviate performance degradation caused by noise.

To summarize, our proposed cSEN addresses the challenges posed by noisy syntax parses in ancient Chinese through a novel two-stage method. Initially, cSEN adopts a Biaffine-attention-based mechanism to measure the confidence of derived syntax parses, which stands out from conventional techniques. These confidence scores actively influence syntax encoding by softly masking noisy arcs and illuminating previously undetected ones. In addition, our incorporation of the Left–Right Branch feature is inspired by the nuanced linguistic characteristics of ancient Chinese—attributes largely neglected by prior research. This feature not only extends the encoding scope of the syntax graph but also adeptly captures the local semantics of the language, enhancing the model's interpretative abilities. Together, these innovations ensure a more robust and accurate representation of ancient Chinese syntax, redressing the prevalent complications tied to harnessing ancient Chinese syntax features.

Table 1
Data statistics of the ancient Chinese poetry thematic classification dataset.

	Train	Dev	Test
#Object-chanting	1129	47	66
#Landscape	1097	44	47
#Persons	2403	91	129
#History	1087	40	76
#Homesickness	9013	357	522
#Mourning	503	18	31
#War	1746	62	115
#Pastoral	1219	47	84
#Farewell	1460	60	83
#Boudoir-plaint	703	34	47
Total	20360	800	1200

4. Experiments

We evaluate the effectiveness of cSEN module using two typical ancient Chinese understanding tasks: thematic classification of ancient poetry and ancient–modern Chinese translation. We build our model by incorporating the cSEN module into existing solid baselines. For the classification task, we follow the work of Vaibhav and Hovy (2019) which has a BERT-GAT-BiLSTM backbone architecture. For the translation task, our model is based on Jin, Wang, and Wan (2020) where dependency graphs are incorporated into neural sequence-to-sequence models with a pointer network.

4.1. Data

To address the scarcity of annotated data for thematic classification, we constructed a novel dataset.[2] Two graduate students specializing in Chinese literature study annotated 22,360 poems, categorizing them into one of ten distinct themes under the guidance of an experienced ancient Chinese linguist.

During the annotation process, we recognized the inherent nature of some poems to embody multiple themes. Given our objective to establish a foundational benchmark for thematic classification, it was imperative to maintain a clear demarcation between themes. Therefore, to eliminate potential ambiguities and to ensure the cohesiveness of our dataset, poems that demonstrated pronounced multi-thematic attributes were filtered out. This measure allowed for a more straightforward classification, facilitating a robust evaluation of our model, especially in its initial stages.

The meticulous annotation process ensured high-quality, reliable annotations. Any conflicted labeling between the two annotators was resolved through consultation with the supervisor, guaranteeing a consistent annotation standard. The dataset is then randomly divided into a training set (20,360), a development set (800), and a test set (1,200). The distribution of themes in the dataset is detailed in Table 1.

For the ancient–modern Chinese translation, we adopt the ancient–modern Chinese parallel corpus contributed by the open-source NiuTrans project.[3] The corpus contains 967,255 sentence pairs extracted from ancient Chinese books. We divided the corpus into training, validation, and test sets with corresponding sizes of 900,000, 60,000, and 7255.

4.2. Syntax parsing

We experiment with two settings — modern supervised parsers and ancient unsupervised syntax derivation. For modern supervised parsing, we adopt the Biaffine dependency parse (Dozat & Manning, 2016) and train it on CTB7 (Xue et al., 2010). For unsupervised syntax derivation, we follow the work of Wu, Chen, Kao, and Liu (2020), which utilizes linguistic knowledge gained from pre-trained language model BERT to infer syntactic dependency structure without direct supervision. We attempt two variants of BERT for syntax derivation and backbone sentence encoder, BERT-wwm-ext (Cui, Che, Liu, Qin, & Yang, 2021) and Anchi-BERT (Tian et al., 2021). BERT-wwm-ext is trained on the modern Chinese corpus containing 5.4B words, while Anchi-BERT is trained upon an ancient Chinese corpus with the size of 39.5M tokens. In addition, we treat the left–right branch as a special kind of syntax parses. Anchi-BERT is trained on a smaller ancient Chinese corpus (39.5M tokens), while BERT-wwm-ext is trained on a larger modern Chinese corpus (5.4B tokens). We also treat left–right branch features as a distinct class of syntax parses.

For clarity, the syntactic parses from the Biaffine parser, BERT-wwm derivation, and Anchi-BERT derivation are denoted as *BiAF*, *WWMD*, *ANCD* respectively, in the following part.

[2] Upon publication of this paper, this dataset will be made available for research purposes.
[3] https://github.com/NiuTrans/Classical-Modern

Table 2
Comparison with baseline model and syntax-aware methods on the thematic classification task. *BiAF*, *WWMD*, *ANCD* denotes parses generated from the Biaffine parser, BERT-wwm derivation, and Anchi-BERT derivation, respectively.

		BERT-wwm		Anchi-BERT	
Methods	Parses	Micro F1	Macro F1	Micro F1	Macro F1
Baseline	None	91.7	89.2	92.4	90.4
GAT	*LRB*	91.5	88.9	93.3	91.4
	BiAF	92.3	89.7	93.3	91.2
	WWMD	91.4	88.8	92.7	90.8
	ANCD	91.8	89.2	93.2	91.0
	BiAF+LRB	92.7	90.4	93.3	91.2
	WWMD+LRB	91.7	89.6	93.2	91.2
	ANCD+LRB	90.8	88.2	92.8	90.7
	BiAF+ANCD+LRB	91.7	88.8	92.6	90.6
cSEN	*BiAF+LRB*	91.4	89.2	93.3	91.6
	WWMD+LRB	92.8	90.7	93.6	**91.9**
	ANCD+LRB	91.3	89.1	93.2	91.3
	BiAF+ANCD+LRB	91.0	89.1	**93.8**	**91.9**

4.3. Implementation and hyper-parameters

For the thematic classification, our model is built by stacking BERT, a graph encoder, and a single-layer LSTM. The graph encoder's node embedding dimension is set to 128, and the hidden size in LSTM is set to 100. We adopt the Adam optimizer with $\rho = 5e-5$ and $\epsilon = 1e-8$, using a batch size of 32. All classifiers are trained for 10 epochs on the train set by default.

We mostly follow the parameter settings from Jin et al. (2020) for the ancient–modern Chinese translation. The Adam optimizer is configured with $\rho = 1e-4$ and $\epsilon = 1e-8$. And all models are trained for 50 epochs with a batch size of 108.

4.4. Results

4.4.1. Ancient poetry thematic classification

Table 2 presents the results of ancient poetry thematic classification. We report the results in Micro-F1 and Macro-F1 scores. The table is divided into three blocks, showing the results of the baseline model, vanilla GAT, and the proposed cSEN. The baseline model achieves 92.4 in Micro F1 and 90.4 in Macro F1, showing strong performance.

From the results in the first two blocks, it can be found that incorporating syntactic trees with GAT encoder brings substantial improvement, proving the value of syntactic information for enhancing ancient Chinese understanding. By comparing the results of employing Anchi-Bert as the sentence encoder and those obtained employing Bert-wwm, we can see that Anchi-Bert outperforms BERT-wwm with a significant lead in all cases. Recall that Anchi-Bert was pre-trained on a much smaller corpus. Also, the performance of syntactic trees derived by BERT-wwm is inferior to the other three. This once more indicates the linguistic gap and syntactic incompatibility between ancient and modern Chinese.

Unsupervised syntax trees derived by Anchi-BERT perform roughly the same as those produced by the Biaffine parser. Additionally, LRB is the best-performing syntax parse among all, improving the performance by 0.9 in Micro F1 and 1.0 in Macro F1. It can be partially explained by the fact that ancient poems are comprised by a few brief sentences, which are highly concise and structurally compact. This results in fewer long-range dependencies, and each token is closely dependent on the immediate preceding or succeeding token.

From the third block, it can be seen that when using Anchi-BERT as sentence encoder, cSEN brings performance gains across all syntax tree setups, raising the top Micro and Macro F1 scores to 93.8 and 91.9, respectively. This demonstrates that: (1) cSEN's denoising capability is effective for utilizing noisy syntactic information to improve ancient Chinese understanding; (2) cSEN can handle noise introduced by different parses, whether it is from a supervised modern Chinese parser or unsupervised derivation.

4.4.2. Ancient–modern Chinese translation

Results of the ancient–modern Chinese translation are shown in Table 3. We use BLEU (Papineni, Roukos, Ward, & Zhu, 2002) and ROUGE (Lin, 2004) scores for performance evaluation.

The baseline model without syntax parses achieves 37.14 in BLEU score and F-scores of 69.71, 46.24, 67.62 in ROUGE-1, ROUGE-2, and ROUGE-L respectively. With single syntactic parses incorporated, all models achieve better performance in all metrics, proving that syntax can effectively improve ancient–modern Chinese translation. LRB is relatively the weakest one, slightly increasing BLEU score by 0.28, and ROUGE f-scores by 0.15, 0.12, 0.10. This might be caused by that sentences from ancient books have more long-distance dependencies and more complicated syntactic structures that left–right branch cannot recover. Anchi-BERT-derived syntax parses have better performance with an improvement of 0.41 in BLEU score, and 0.19, 0.29, and 0.23 in ROUGE scores. BERT-wwm derived syntax trees and trees generated by Biaffine parser have similar results. In contrast to Anchi-BERT derived trees, their performance is inferior in BLEU scores but better in ROUGE F-scores. Feeding multiple syntactic parses into the GAT-based

Table 3
Experimental Results of the ancient and modern Chinese translation task.

Methods	Parses	BLEU	RG-1 F-score	RG-2 F-score	RG-L F-score
Baseline	None	37.14	69.71	46.24	67.62
GAT	*LRB*	37.42	69.86	46.36	67.72
	BiAF	37.45	70.23	46.93	68.21
	WWMD	37.46	70.20	46.89	68.14
	ANCD	37.55	69.90	46.53	67.85
	BiAF+ANCD+LRB	34.62	69.20	45.15	67.15
cSEN	*BiAF+ANDC+LRB*	**37.73**	**70.27**	**47.09**	**68.23**

Table 4
Ancient-to-modern Chinese translation examples generated by the baseline model, vanilla GAT, and cSEN. The first block shows the original ancient Chinese sentence (src), human-annotated modern Chinese reference (ref), and corresponding English translations (trl).

Sent	Ancient Chinese
1	src: 父岘，少骁勇，事章大宝为帐内部曲。 ref: 父亲陈岘，年轻时骁勇，侍奉章大宝，任帐内部曲。 trl: His father, Chen Xian, was brave when he was young. He served Zhang Dabao, as an in-tent soldier.
2	src: 晋伐骊戎，骊戎男女以骊姬。归生奚齐。其娣生卓子。 ref: 晋国攻打骊戎，骊戎男把骊姬献给晋献公，回国后生了奚齐，她的妹妹生了卓子。 trl: Jin state attacked Li Rong. Li Rong's men presented Li Ji to Jin Xian Gong. After returning the state, Li Ji gave birth to Xi Qi, and her sister gave birth to Zhuo Zi.
3	src: 及竣事，又曰：杨存中之罢，朕不安寝者三夕。 ref: 等到事毕后，又说：杨存中被罢免，我有三天不得安寝。 trl: After the matter was over, he said: "The dismissal of Yang Cunzhong, made me unable to sleep for three days."
Baseline	
1	父亲张岘，少年时骁勇善战，在章大宝帐下当兵, 任章大宝的部曲。 His father, Zhang Xian, was brave and good at fighting when he was young. He served as a soldier under the command of Zhang Dabao, and was Zhang Dabao's soldier.
2	晋国讨伐骊戎，骊戎男女献上骊姬，回国后生了奚齐，骊戎的妹妹生了卓子。 Jin state attacked Li Rong, and Li Rong's men and women presented Li Ji, and after returning the state, they gave birth to Xi Qi, and Li Rong's sister gave birth to Zhuo Zi.
3	等到完成任务，又说：杨存中被罢免，朕三夜不安寝。 After completing the task, he said: "The dismissal of Yang Cunzhong, made me unable to sleep for three nights."
Vanilla GAT	
1	父亲李岘，少年时骁勇善战，在章大宝手下做事。 His father, Li Xian, was brave and good at fighting when he was young, and worked under Zhang Dabao.
2	晋国讨伐骊戎，骊戎男娶了骊姬，嫁给了奚齐，奚齐的妹妹生了卓子。 Jin state attacked Li Rong. Li Rong's men married Li Ji, married Xi Qi, and Xi Qi's sister gave birth to Zhuo Zi.
3	等到完工，又说：杨存中被罢免的事，我感到不安宁 的有三个晚上。 After the work was finished, he said: I felt uneasy about Yang Cunzhong's dismissal for three nights.
cSEN	
1	父亲李岘，年轻时骁勇，事奉章大宝任帐内部曲。 His father, Li Xian, was brave when he was young. He served Zhang Dabao as an in-tent soldier.
2	晋国攻打骊戎，骊戎男把骊姬送给晋国，回国后生了奚齐，她的妹妹生了卓子。 Jin state attacked Li Rong. Li Rong's men presented Li Ji to Jin State. After returning the state, Li Ji gave birth to Xi Qi, and her sister gave birth to Zhuo Zi.
3	等到事情完毕，又说：杨存中被罢免，我三天不安寝。 After the matter was over, he said: "The dismissal of Yang Cunzhong, made me unable to sleep for three days."

model simultaneously leads to a significant performance drop. While replacing GAT with the proposed cSEN increases performance in all matrices, with 37.73 in BLEU score and 70.27, 47.09, 68.23 in ROUGE F-scores. From the above results, we conclude that syntax parses from unsupervised derivation or modern Chinese syntax parsers introduce noise and degrade model performance. With our confidence learning, model is able to distinguish and separate informative syntactic information from noise, thus alleviating its negative effect.

Table 4 shows three ancient-to-modern Chinese translation examples produced by different models. From generations for Sent 1, we can see a common error: due to the lack of contextual information, all three models assume the surname of "the father" using the most common Chinese surnames, such as "Li" and "Zhang". For Sent 2, the generations from the baseline model and vanilla GAT differ significantly from the human-annotated reference. They fail to recognize the relationship between the characters, such as who "其娣" refers to, thus generating translations that do not correspond to the facts. In contrast, with stronger denoising capability, cSEN is able to correctly encode the information in ancient Chinese texts, thus producing higher-quality translations.

Table 5
Ablation study results.

Variants	Micro F1	Macro F1
cSEN	**93.8**	**91.9**
w/o Confidence	92.8	91.1
w/o Gate	93.0	91.0

Table 6
Comparison of different combination configurations on syntactic parses. Parses in square brackets are merged onto a single adjacency matrix and parses in parentheses are incorporated by the gated mechanism.

Syntax trees	Micro F1	Macro F1
[*ANCD*] + (*LRB*)	93.2	91.3
[*BiAF*] + (*LRB*)	93.3	91.6
[*BiAF* + *LRB*] + (*ANCD*)	92.8	90.9
[*ANCD* + *LRB*] + (*BiAF*)	92.8	90.5
[*BiAF* + *ANCD*] + (*LRB*)	**93.8**	**91.9**

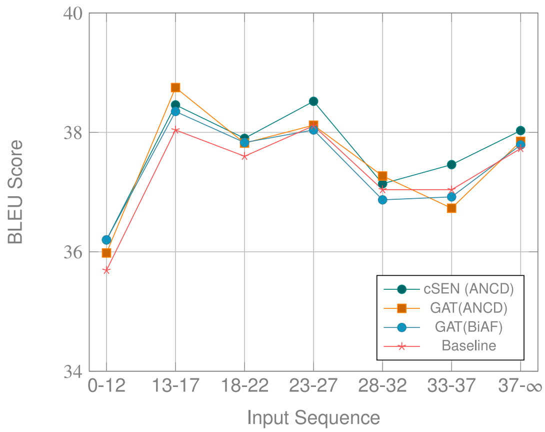

Fig. 3. BLEU scores for different input sentence lengths.

4.5. Exploration

In this section, we investigate the impact of different cSEN components and analyze the nature of different syntax parses. First, we conduct ablation studies on cSEN. Results are reported in Table 5. Both the removal of the confidence (equivalent to vanilla GAT) and the removal of the gated mechanism lead to significant performance degradation. This indicates that both modules are crucial for effectively utilizing syntactic information from noisy parses.

Second, we experiment with different syntax tree combination setups. Table 6 lists the findings. Combining two syntax parsers onto a single dependency graph can provide richer syntactic information and produce higher performance than using alone syntax parses, no matter if it is from unsupervised derivation or a supervised parser. We also explore the incorporation position of LRB features. It can be observed that the model performance suffers significantly if LRB features and graph parses are directly merged together. This again indicates the necessity of our gated method for LRB feature integration.

Third, as illustrated in Fig. 3, we compare our model and baselines over different input lengths. cSEN performs better in relatively longer sentences, according to the results. This supports the hypothesis that syntax helps guide longer sentence understanding as dependency reduces the distance. Because of the incompatibility between modern and ancient Chinese, unsupervised derivation is more effective than supervised parsing when compared to other syntax parsers. In most cases, cSEN yields better performance due to its stronger denoising capabilities.

5. Conclusions

In this paper, we investigate the role of syntax in improving ancient Chinese understanding. Due to a lack of syntax annotation, syntax trees are obtained by unsupervised derivation and supervised modern Chinese parsers. To alleviate the negative effect of noise, we propose a confidence-based syntax encoding network (cSEN). Experimental results on two typical ancient Chinese understanding tasks show that our model can effectively distinguish informative syntactic information from noise and achieve better performance.

This work has multiple implications for ancient Chinese Understanding and related studies. Theoretically, the development of cSEN brings a new perspective to NLP models dealing with noise in data. This model, which uses Biaffine transformations and a left–right branch (LRB) to capture local features, illustrates how syntax parsing noise can be mitigated to improve language understanding. In addition, the research highlights how ancient Chinese linguistic characteristics can be taken into consideration to build more sophisticated NLP models, providing a blueprint for future studies focusing on other ancient or low-resource languages. Practically, the research findings show significant improvements in the ancient–modern Chinese translation task and in the ancient poetry thematic classification tasks. Researchers can leverage this model to create more accurate translations and classifications, which are foundational tasks in ancient Chinese research. More importantly, our work can serve as a useful resource in the education about the unique syntax of ancient Chinese and how it differs from that of modern Chinese.

While our research makes significant strides in improving the understanding of ancient Chinese through novel NLP techniques, there are still some limitations to consider. The principal limitation stems from the increased computational cost associated with the calculation of confidence, which necessitates two distinct self-attention operations along with a Biaffine transformation. This multiplication of parameters inadvertently leads to a more laborious training process and necessitates increased computational storage, potentially constraining the application of our model in environments with limited resources. In light of these constraints, our future work is oriented towards mitigating this limitation by focusing on optimizing the computational requirements of our model.

CRediT authorship contribution statement

Shitou Zhang: Formal analysis, Methodology, Visualization, Writing – original draft, Writing – review & editing. **Ping Wang:** Conceptualization, Funding acquisition, Supervision, Writing – original draft. **Zuchao Li:** Conceptualization, Data curation, Methodology, Project administration, Visualization, Writing – review & editing. **Jingrui Hou:** Writing – original draft, Writing – review & editing. **Qibiao Hu:** Writing – original draft, Writing – review & editing.

Data availability

Data will be made available on request.

Acknowledgments

This work was supported by the National Natural Science Foundation of China [No. 72074171 and No. 72374161] and the China Scholarship Council (CSC) [No. 202208060371].

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